

Article

Artificial Intelligence Technologies as Smart Solutions for Sustainable Protected Areas Management

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Abstract: Artificial intelligence (AI) is becoming not only an auxiliary tool, but also one of the main factors helping to shape natural resource management models. The application of artificial intelligence in protected areas allows for a transition to more sustainable management of protected areas. By applying artificial intelligence technologies, it is possible not only to respond to changes or violations that have already occurred but also to more effectively predict potential threats, form long-term protection strategies, and make rational decisions based on accurate and timely data analysis. This study aims to determine the possibilities and importance of applying artificial intelligence technologies to the sustainable management of protected areas. The sample group of this study consists of a total of 135 experts from Turkey, Lithuania, and Morocco (45 from each country). The sample includes professionals with expertise in the relevant field, namely lawyers (9), academics (9), managers of protected areas (9), government officials responsible for protected areas (9), and representatives of non-governmental organizations (9). This study employed qualitative research methods, within which a case study design was adopted. For the analysis of the findings, thematic analysis and content analysis techniques were utilized to ensure a comprehensive and in-depth interpretation of the data. Analysis of the results of this study showed that integrating AI into the management of protected areas increases management efficiency and helps create long-term strategies, but successful application depends on cooperation between technology developers, scientists, and environmental specialists. Also, AI applications are expected to be a critical part of the process of environmental sustainability and fighting climate change.

Keywords: artificial intelligence (AI); protected areas; sustainable management; smart solutions



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1. Introduction

Currently, artificial intelligence (AI) is gaining increasing importance and establishing itself as a strategic tool in various fields, including the environmental field [1–4]. With the rapid development of technologies and the increasing of environmental challenges, artificial intelligence (AI) is starting to play an increasingly important role in the sustainable management of natural resources [5–8]. These technologies are changing not only industries or everyday life but also becoming an important tool in preserving biodiversity, reducing the impact of human activities on the environment, and creating advanced, science-based management systems [9,10]. Smart solutions that allow for faster, more efficient, and more accurate monitoring and analysis of environmental changes are needed. Smart solutions

refer to applications that integrate technological innovations and data-driven approaches to enhance the effectiveness and sustainability of complex management processes [11,12]. In the context of protected area management, these solutions involve the collection, analysis, and integration of environmental data into decision-support systems. Artificial intelligence lies at the core of this framework, optimizing decision-making processes in key areas such as monitoring, enforcement, and resource management through technologies like machine learning, image processing, and natural language processing [4,7]. In this regard, AI-based applications emerge as one of the fundamental components of smart solutions used in protected areas.

Artificial intelligence technologies, using large amounts of data, sensors, remote sensing, and learning algorithms, can recognize complex patterns in natural processes, detect threats in real time and predict possible changes in ecosystems. In this way, AI becomes not only an auxiliary tool, but also a strategic decision-making tool in the context of sustainable development [13,14]. Increasingly, managers of protected areas, scientists, and policymakers are starting to apply AI technologies to ensure effective conservation of natural heritage and a balanced relationship between human activity and the natural environment [15]. The application of AI allows for faster and more accurate data processing, since monitoring and analyzing protected areas requires a large volume of data—from meteorological indicators to animal migration or habitat changes [16,17]. AI can quickly process this data, detect patterns, and provide insights that would be difficult to achieve using traditional methods [18,19]. Another important point is that AI can predict certain phenomena—for example, the probability of forest fires, the spread of invasive species, or the impact of climate change on habitats [20,21]. This allows for preventive measures and timely responses. Since the field of environmental protection often faces limited financial and human resources, AI helps optimize work—for example, automating the monitoring of territories with drones or satellites, reducing the need for physical patrols, and identifying violations more quickly [22,23].

Artificial intelligence (AI) applications can provide significant contributions in protected areas by enabling rapid data collection and processing, surpassing traditional methods, and facilitating swift outcomes [18]. Through monitoring systems that gather and analyze large datasets, AI can ease the detection of adverse environmental impacts. Leveraging big data, predictive modeling can support preventive measures in areas such as forest fires, illegal activities detection, and similar environmental threats. Furthermore, AI plays a decisive role in optimizing resource utilization to achieve maximum efficiency [21,23]. Artificial intelligence (AI) technologies and applications are increasingly preferred across various sectors, primarily due to the operational convenience they offer and their positive contributions to management processes. In the context of the sustainable management of protected areas, the integration of AI technologies is not only inevitable but also essential [6]. These areas are typically vast and subject to significant human activity, which necessitates advanced and efficient management mechanisms. Accordingly, the facilitation of management processes through AI-supported systems and the transfer of relevant technologies can pave the way for more effective and sustainable governance. In this regard, AI technologies should be regarded not merely as a preference but as a strategic necessity. Their implementation can significantly contribute to the achievement of environmental sustainability goals, mitigate environmental degradation, and reduce risks to biodiversity.

There is a growing body of research on the use of artificial intelligence (AI) in the management of protected areas, with a particular focus on biodiversity conservation, early detection of wildfires, and monitoring of illegal hunting activities. However, these studies often lack a holistic and systematic perspective and tend to overlook the integration of AI into administrative and governance processes. Unlike existing studies, this research

not only addresses the technical use of AI within structural and institutional management frameworks, but also provides practical, application-oriented analyses. Drawing on expert opinions relevant to the research topic, this study explores ways to enhance the capacity of AI in protected area management and offers theoretical insights into its integration into governance processes. This study stands out by providing a clear idea of how AI can be used in managing protected areas and showing how it can support sustainable management.

This study aims to determine the possibilities and importance of applying artificial intelligence technologies to the sustainable management of protected areas, assessing their benefits, areas of application, and challenges, to ensure effective protection of the natural environment and long-term preservation of ecosystems.

2. Materials and Methods

In this study, qualitative research methods, which are frequently employed in social sciences research, were utilized. Specifically, a case study design was adopted within the scope of qualitative research methods. Qualitative research methods provide researchers with opportunities to move from parts to the whole and to conduct a holistic analysis of cases [24,25]. Moreover, these methods offer an in-depth perspective aimed at understanding the complexity and multidimensional nature of the phenomenon under investigation. Unlike quantitative approaches that focus on numerical data, qualitative methods prioritize participants' expressions, experiences, and meanings. This enables researchers to interpret behaviors, attitudes, and social dynamics of individuals and groups in a more comprehensive and contextualized manner, thereby facilitating a deeper comprehension of the essence and context of the phenomenon [26–28].

The case study design is a qualitative research method that facilitates the in-depth examination of a specific event, phenomenon, individual, group, or institution. This design enables researchers to analyze a phenomenon holistically by considering contextual factors [29,30]. Case studies are typically conducted using various data collection techniques (such as interviews, observations, and document analysis) and aim to generate meaning through an inductive approach. In this study, a semi-structured interview form was employed with experts to obtain in-depth information. This method was particularly preferred for understanding the complex situations in three countries with distinct characteristics and for conducting a comparative analysis of these contexts. Widely used in numerous social science studies, this approach facilitates the identification and interpretation of common themes and sub-themes [29] and this method serves as a valuable tool for contributing to theoretical inferences and explaining complex processes [31,32].

Sample Group

The sample group of this study consists of a total of 135 experts from Turkey, Lithuania, and Morocco (45 from each country). The sample includes professionals with expertise in the relevant field, including lawyers (9), academics (9), managers of protected areas (9), government officials responsible for protected areas (9), and representatives of non-governmental organizations (9). To obtain in-depth insights, a purposive sampling method was used in the selection of the sample group. To ensure the confidentiality of the sample group included in this study, no identifying information about the participants was included. All participants were appropriately coded based on their country of origin. Quotations were reported without referencing any personal or institutional information. The interview data obtained were securely stored by the researchers. In addition, an informed consent form was used prior to the interviews. These procedures were meticulously implemented to ensure full compliance with ethical standards and the requirements of research ethics. Careful attention to participants' privacy enhances the reliability and validity of this study, while simultaneously enabling participants to express their views freely and

candidly. Consequently, the quality of the data obtained and the ethical integrity of the research have been duly preserved.

The inclusion of lawyers, academics, managers, government officials, and NGO representatives in the research group was guided by their strategic and significant roles in the sustainable management of protected areas. Lawyers possess domain-specific expertise regarding the potential legal implications and ethical violations that may arise from the use of AI technologies. Academics, particularly in recent years, have been conducting research focused on the diverse applications of AI across various fields. Government officials working in protected areas are considered direct stakeholders in both the problems and solutions associated with these contexts. NGO representatives act as vital intermediaries between the public and institutions, reflecting the perspectives and participation of local communities in addressing relevant challenges. Therefore, the sample group was selected with the rationale that these participants could provide in-depth knowledge aligned with the research questions and proposed solutions.

The names of the participants, the institutions they work for, and their locations have been kept confidential by the researchers. Coding was used to present expert opinions, and the coding scheme for experts from each country is provided in Table 1 below:

Table 1. Sample group and coding scheme.

Country	Experts	Codes
Turkey	Lawyer	LT (1-2-3-4-5-6-7-8-9)
	Academics	AT (1-2-3-4-5-6-7-8-9)
	Forest Managers	FMT (1-2-3-4-5-6-7-8-9)
	Forest Expert from Government	FET (1-2-3-4-5-6-7-8-9)
	Non-governmental organization Expert (NGO)	NGOET (1-2-3-4-5-6-7-8-9)
Lithuania	Lawyer	LL (1-2-3-4-5-6-7-8-9)
	Academics	AL (1-2-3-4-5-6-7-8-9)
	Forest Managers	FML (1-2-3-4-5-6-7-8-9)
	Forest Expert from Government	FEL (1-2-3-4-5-6-7-8-9)
	Non-governmental organization Expert (NGO)	NGOEL (1-2-3-4-5-6-7-8-9)
Morocco	Lawyer	LM (1-2-3-4-5-6-7-8-9)
	Academics	AM (1-2-3-4-5-6-7-8-9)
	Forest Managers	FMM (1-2-3-4-5-6-7-8-9)
	Forest Expert from Government	FEM (1-2-3-4-5-6-7-8-9)
	Non-governmental organization Expert (NGO)	NGOEM (1-2-3-4-5-6-7-8-9)

The purposive sampling method is a sampling technique based on the deliberate selection of individuals/experts who possess the most relevant knowledge to answer a specific research question. In this method, data sources are selected according to predefined criteria rather than randomization [33]. Widely used in qualitative research, purposive sampling aims to obtain in-depth information and explore context-specific details. This method has various types, and it enables researchers to create a sample that is well suited to the research topic and has a high level of representativeness, thereby enhancing the validity and reliability of this study [34–36]. In accordance with the principles of the purposive sampling method, the sample group was limited to a total of 135 participants from Turkey, Lithuania, and Morocco (45 from each country). Prior to and during this study, the participants were provided with the necessary information about the research, notified of the interview dates, and the required interviews were conducted.

In this study, data saturation was rigorously pursued. To achieve this, recurring responses from participants were thematically analyzed, and common patterns were identified [37]. In cases where responses hindered the emergence of consistent themes, interviews were either concluded or redirected to the next question to maintain thematic coherence. To ensure inter-coder reliability across the datasets and interview groups, independent coding procedures were applied, aiming for a high level of agreement among coders [38].

Furthermore, several measures were taken to minimize researcher bias, including reflexive documentation of interview records and the implementation of the constant comparative method throughout the analysis process. These approaches have not only enhanced the objectivity of the data but also strengthened the consistency and reliability of the findings. Consequently, the scientific validity of this study has been duly supported [39].

The selection of Turkey, Lithuania, and Morocco for this study is based on the similarities and contrasts in the environmental policies, technological adoption, and management practices of protected areas in these countries. In Turkey, environmental issues have been on the rise due to the increasing population and industrialization. However, significant strides have been made in recent years. Initiatives for using AI technologies, particularly in combating forest fires and natural disasters, have been implemented, but issues such as public participation and infrastructure remain as barriers [40–42].

Lithuania, as a member of the European Union, possesses comprehensive legal frameworks for environmental protection. However, resistance has been encountered in the process of biodiversity conservation and combating climate change. Efforts are focused on breaking this resistance through collaborations, and the importance of AI technologies is emphasized, along with encouraging public participation [43,44].

Morocco, as a developing country, has made significant progress in environmental sustainability and combating climate change. However, challenges such as inadequate legal frameworks and infrastructure deficiencies remain prominent. Morocco is actively working on strengthening local participation, updating legal regulations, and integrating AI technologies in its environmental policies and practices [45–47].

Data Collection Tool

To conduct interviews with the targeted sample group, semi-structured interviews were used as the primary data collection method. For this purpose, a structured interview form was developed by researchers. The preparation of this form was based on a comprehensive literature review, and it was reviewed by experts in the field as well as language specialists.

A semi-structured interview form serves as a data collection tool that allows researchers to gather in-depth information from participants within the framework of predetermined key questions [48]. Although the form includes open-ended questions prepared in advance, it also allows researchers to exercise flexibility during the interview process by posing additional questions and elaborating on responses when necessary. The interview form was developed based on the opinions of subject matter experts, ensuring content validity. Additionally, the interview form was tested with pilot groups prior to this study and revised based on feedback from pilot participants to reach its final version. Specifically, the questions were refined to enhance the data collection capacity, ensuring the acquisition of accurate and reliable data for this study [49]. This method broadens the scope of the research by enabling a more in-depth analysis of participants' experiences, thoughts, and emotions. Pilot implementations are of considerable importance in assessing whether the research questions effectively serve the purpose of this study. A well-constructed set of questions facilitates the collection of reliable and in-depth information in line with the research objectives. Moreover, redundant or repetitive questions are removed from the interview form to ensure that participants are guided accurately and efficiently throughout the process [50]. The questions included in the interview form used in this study are as follows:

1. What are the main problems and challenges in achieving sustainability in protected areas management?
2. What do you think about the applicability and current use of AI technologies in protected areas management? Which AI applications could be particularly effective?

3. What is the potential of AI in achieving sustainable protected areas management goals in your country? How can these technologies contribute to environmental sustainability?
4. What are the main legal challenges related to the application of artificial intelligence in protected areas management? What legal measures can encourage the implementation of AI in sustainable forest management?
5. Does the use of AI technologies in forest management create any challenges in terms of legal frameworks and ethical standards in Morocco? How is regulation and oversight applied in this?
6. What collaboration models or strategies would you suggest for public institutions to adopt AI technologies in forest management? What is the status and future of such collaborations?
7. How do you see the role of AI technologies in protected areas management evolving in the future? What strategic plans do in your country have to implement these technologies on a larger scale?

Analysis of Interview Data

The data obtained from interviews conducted with a sample group of 135 experts from both countries were processed and analyzed using the Nvivo 14 software. The data were transformed into findings through descriptive analysis and content analysis. Since the emphasis in interview analyses and content evaluations is on the participants' statements rather than numerical expressions, the responses of the managers in the interview group were directly quoted [51].

3. Results

In this section of this study, the findings obtained through interviews conducted in line with the research objectives are presented. Researchers summarized the structured interview responses in the table below and included direct quotes from participants to ensure transparency and provide deeper insights. Additionally, to gain an in-depth understanding of the subject, participants' opinions and perspectives were directly quoted in accordance with the principle of transparency in data sharing.

Legal professionals, academics, forest area managers, government officials, and experts from non-governmental organizations in all three countries were asked the following question: *"What are the main problems and challenges in achieving sustainability in protected areas management?"* The responses provided by experts from each country were analyzed using content analysis and evaluated accordingly. The results are presented in Table 2 below.

Table 2. Sustainability challenges in the management of protected areas in Turkey, Lithuania, and Morocco.

Challenge Area	Turkey	Lithuania	Morocco
Economic and Social Activities Conflicting with Conservation	Local communities resistant to sustainable management; dependence on forest products.	Agriculture; forestry conflict with nature protection.	Tourism, agriculture, and conservation clash; land use conflicts.
Climate Change and Natural Threats	Forest fires; climate change, over-exploitation.	Climate change impacts species habitats.	Climate change affecting forest regeneration.
Limited Resources and Infrastructure	Limited budgets; lack of scientific data collection resources.	Limited budgets for protected area management; infrastructure difficulties.	Limited technological infrastructure; insufficient funding for monitoring.

Table 2. Cont.

Challenge Area	Turkey	Lithuania	Morocco
Tourism and Human Activities Environmental Impacts	Excessive tourism; recreational activities causing ecosystem damage.	Increased visitor traffic causing littering, noise, and ecosystem damage.	Tourism and environmental degradation; conflicting demands.
Biodiversity and Illegal Activities	Illegal hunting; logging; weak deterrent penalties.	Monoculture forestry reducing biodiversity; invasive species.	Deforestation; illegal logging; impact on biodiversity.
Inadequate Legal Frameworks and Conflicting Laws	Inadequate local laws; insufficient enforcement of regulations.	Conflicting regulations; lack of cooperation between institutions.	Conflicting land use demands; lack of real-time data for decision making.
Local Community Participation	Insufficient local community involvement in conservation.	Limited involvement of local communities; insufficient education programs.	Lack of community involvement in sustainable management.

Based on expert opinions obtained in this study, the challenges encountered in the three countries are categorized under seven (7) main themes: “Economic and Social Activities Conflicting with Conservation”, “Climate Change and Natural Threats”, “Limited Resources and Infrastructure”, “Tourism and Human Activities’ Environmental Impacts”, “Biodiversity and Illegal Activities”, “Inadequate Legal Frameworks and Conflicting Laws”, and “Local Community Participation”.

In the context of the emerging theme “Economic and Social Activities Conflicting with Conservation”, FMM-4 highlights the relationship between financial development and environmental conservation in the country, emphasizing that economic activities may be disrupted if the natural environment deteriorates (AT-2). Under the heading “Climate Change and Natural Threats”, it is emphasized that global climate change increases risks in protected areas and negatively affects the conservation strategies of local managers (NGOEL-3). Another threat identified is under the theme “Biodiversity and Illegal Activities”, where it is pointed out that illegal hunting activities pose a significant risk to biodiversity. Illegal hunting represents a major threat to our country, with people hunting certain species despite the risk of extinction. This is highlighted as a significant issue for biodiversity (LM-6). Another concern is the “Local Community Participation” theme, where the lack of local community involvement in the sustainable management of protected areas is considered one of the biggest obstacles. The sustainable management of protected areas is only possible when the local population feels a sense of ownership over the region. Social participation is essential and must be effective (FML-4).

When comparing the challenges faced by the three countries, the most prominent difficulties in Turkey and Lithuania are identified as deficiencies in legal regulations and the lack of local community participation.

Sustainability in protected areas requires the effective implementation of legal frameworks. However, in many cases, existing laws are not sufficiently robust, and local authorities fail to enforce regulatory mechanisms effectively (LT3-TNGOE1). Conflicting legislation and regulatory inconsistencies create legal uncertainty, limiting sustainable conservation efforts and complicating collaboration among relevant stakeholders. These legal inconsistencies hinder the effective implementation of conservation policies (LL7-LA4). Another major challenge in achieving sustainable management of protected areas is the reluctance of local communities to adopt and implement conservation practices (FMT6). Sustainable

management practices in protected areas face many obstacles. The most important of these is that the people living in that area contribute little to the conservation of these areas and hinder this process. Therefore, the sustainable management approach is sometimes disrupted (FML9).

In Morocco, there is a lack of digital infrastructure in the management of protected areas, and environmental crimes and illegal activities are quite common. However, Turkey, Lithuania, and Morocco face common challenges such as declining biodiversity, climate change and environmental degradation, and intensification in the tourism sector.

The limited availability of financial resources and the slow pace of technological advancements disrupt effective monitoring processes in protected areas, thus hindering sustainable management. The lack of technological infrastructure negatively impacts data collection and analysis, preventing the timely detection of environmental changes and the implementation of appropriate measures (LM4, FEM5-LM3).

Experts from all three countries were asked the following question: “What do you think about the applicability and current use of AI technologies in protected areas management? Which AI applications could be particularly effective?” The responses were analyzed using content analysis, and the findings are presented in Table 3 below.

Table 3. AI applications and their purposes in protected areas management in Turkey, Morocco, and Lithuania.

Country	Current State of AI Applications	Effective AI Applications	Infrastructure and Legal Barriers	Education and Local Involvement
Turkey	AI applications have started to develop in Turkey, but there are infrastructure gaps and legal barriers.	Smart monitoring systems Forest fire detection Monitoring illegal activities	Infrastructure deficiencies Data security and legal regulations Legal barriers to monitoring illegal activities	Education of local communities and authorities is required Collaboration between public and private sectors is essential
Lithuania	AI applications are not widely utilized, but early steps have been taken.	Satellite data and GIS technologies Drone use Smart sensor networks chatbots and visitor flow management	Infrastructure gaps and legal regulations Need for local authorities to adapt to the technology	Training for local authorities and raising awareness in communities is necessary
Morocco	AI technologies are in the early stages of adoption, with limited use in forest management.	Remote sensing (satellite imagery) Drone use and IoT sensor networks Predictive modeling (forest fire risk prediction)	Lack of infrastructure investment Insufficient training at the local level	Training is necessary Encouraging collaboration between government bodies and the private sector

Based on expert opinions obtained in this study, responses regarding the potential use of artificial intelligence (AI) and the specific AI tools that could be employed in protected areas management across the three countries have been categorized into four (4) main themes: “Current State of AI Applications”, “Effective AI Applications”, “Infrastructure and Legal Barriers”, and “Education and Local Involvement”.

Although AI technologies hold significant potential for the management of protected areas in all three countries, each faces different challenges and operates at varying levels

of implementation. Turkey, Lithuania, and Morocco require investments in infrastructure, education, and local-level collaboration to effectively integrate these technologies. In Turkey, legal barriers and infrastructure deficiencies are the primary obstacles, whereas in Morocco and Lithuania, in addition to infrastructure gaps, the training of local authorities and communities is crucial. The successful implementation of AI technologies in the management of protected areas critically depends on the participation and education of local communities. Enhancing the technological literacy of local stakeholders can serve as a key objective to improve their basic understanding of AI applications. In doing so, particular attention must be paid to legal and ethical concerns, while also enriching the everyday experiences of both decision-makers and participants. Moreover, the use of AI technologies should be adapted to the socio-cultural differences among Turkey, Lithuania, and Morocco. Potential disadvantages that may arise from these contextual differences should be carefully avoided.

AI can be particularly beneficial for monitoring natural disasters (e.g., wildfires) and enabling rapid response mechanisms. However, the successful implementation of these technologies requires collaboration at both local and national levels as well as investment in infrastructure. Ensuring the sustainability of AI-based technologies also necessitates regular maintenance and technical support (FMT4-6). In Lithuania, AI has the potential to make significant contributions to nature conservation, but this would require investments in data collection, adaptation of AI models to local conditions, and ensuring that institutions and experts are adequately prepared to utilize these technologies (FML2-AL3-7). Strengthening data collection mechanisms, enhancing technological infrastructure, and developing localized solutions can enable the effective use of AI and other advanced technologies, thereby generating both environmental and economic benefits (FEM5).

Furthermore, the existing AI tools and applications used in the three countries have been categorized into five (5) main areas: “Remote Sensing/Satellite Imagery”, “Drones”, “Smart Monitoring Systems”, and “Predictive Modeling”. These AI technologies are primarily used for forest fire prevention, biodiversity conservation, and combating illegal activities in all three countries. A close examination of the research findings reveals that, particularly, “Smart Monitoring Systems” and “Predictive Modeling” methods stand out. Smart monitoring systems can be considered highly effective in tracking and monitoring changes that may occur in the natural environment. These systems primarily collect data through satellite data and camera networks, providing ease of use for users. Through real-time data, they can play a decisive role, especially in monitoring ecosystems and biodiversity. Predictive modeling systems can play a critical role in identifying potential threat risks. They can be regarded as valuable tools, particularly in the early detection and suppression of forest fires and in detecting illegal activities. For instance, past fire incidents and their causes in forested areas can be taught to the system using machine learning methods, which can help reduce potential risks.

Experts from all three countries were asked the following question: “What is the potential of AI in achieving sustainable protected areas management goals in your country? How can these technologies contribute to environmental sustainability?” The responses were analyzed using content analysis, and the findings are presented in Table 4 below.

Based on the opinions of experts involved in the research, the purposes of using artificial intelligence tools in three countries are grouped under seven (7) main categories. These categories are as follows: “Species Identification”, “Forest Ecosystem Monitoring”, “Illegal Activity Detection”, “Climate Change Analysis”, “Resource Management”, “AI Technologies Used”, and “Level of AI Development”.

Table 4. AI potential and its contribution to environmental sustainability in protected areas management in Turkey, Morocco, and Lithuania.

AI Application Areas	Turkey	Morocco	Lithuania
Species Identification	Monitoring biodiversity and environmental crimes	Track wildlife in protected areas	Automatic animal species recognition
Forest Ecosystem Monitoring	Forest health, fire detection, and ecosystem monitoring	Drones and IoT sensors for ecosystem health tracking	Drones and sensors for monitoring forest health and animal populations
Illegal Activity Detection	AI used for detecting illegal logging, poaching, and other environmental crimes	AI used in surveillance to combat illegal activities	PAWS AI for predicting poaching locations; TrailGuard AI for real-time monitoring
Climate Change Analysis	AI supports monitoring climate effects on forests and biodiversity	AI helps mitigate environmental impacts through climate adaptation strategies	AI for climate change adaptation and ecosystem monitoring
Resource Management	Data-driven decision making for sustainable forest management	AI in water and soil management for better land-use planning	GIS and satellite data for monitoring protected areas (not fully integrated with AI)
AI Technologies Used	Satellite imagery, drones, sensors, and data analysis	Remote sensing, drones, and predictive modeling	GIS, drones, sensors, satellite imagery, and mobile guides
Level of AI Development	Advanced use, particularly in forest and biodiversity management	Early-stage AI applications	Emerging AI use with significant growth potential

The use of artificial intelligence (AI) technologies in the sustainable management of protected areas in Turkey is becoming increasingly widespread. Turkish experts state that AI applications are being utilized in monitoring biodiversity, tracking environmental crimes, and combating forest fires.

Today, artificial intelligence technologies are an increasingly widespread and powerful tool in the sustainable management of protected areas. It is an important tool for the protection and sustainability of biodiversity and combating forest fires, which have recently increased, and it also makes significant contributions to the protection of environmental health (FMT5-FET6.8). It also contributes significantly to the detection and reduction of environmental crimes. Facilitating management processes and effective management is a very important process (LT1-7).

While it is stated that the use of AI in protected areas in Morocco is still in its early stages, it appears that AI technology is mostly utilized in the tracking of environmental crimes and natural resource management. Moroccan experts state that AI-powered drones and sensors are preferred for this purpose.

Although the use of AI in the management processes of protected areas in Morocco is not yet widespread enough, there are initiatives to apply AI, especially in the identification and monitoring of different species and, of course, in the fight against forest fires (FEM3-4-5). AI is quite popular for biodiversity conservation and sustainable management of natural areas, but yes, it is not yet very widespread. We need to overcome this situation and increase the use of AI in management processes. In the process, AI infrastructure investments should be increased, and we should build a strong technology infrastructure for the future (NGOEM2-9).

Lithuania is not yet at the desired level in the use of AI compared to Turkey and Morocco, and it is acknowledged that the integration process is ongoing. It is evident that biodiversity is mostly monitored through satellite data, and environmental crimes are monitored. Although the use of AI technologies in Lithuania has not yet become widespread, it is understood that tools such as PAWS AI (Protection Assistant for Wildlife Security) and TrailGuard AI are utilized particularly for monitoring wildlife areas. PAWS AI and TrailGuard AI represent rapidly advancing technological tools within the field of artificial intelligence. These systems are primarily employed for detecting illegal hunting activities and supporting the protection of wildlife in natural habitats. Notably, they are considered highly effective in safeguarding biodiversity within wildlife conservation areas.

Lithuania. No, the use of AI in the management of protected areas is not yet at the desired level, there is actually a huge potential, but as I said, there is not yet widespread use, unfortunately (NGOEL5). There is a huge transformation, and it is necessary to support the use of AI in the management of protected areas. The use of AI is inevitable, especially for monitoring and protecting environmental health, tracking and reducing environmental crimes, and harmonizing tourism activities in these areas with the environment (FEL5).

Experts from all three countries were asked the following question: “What are the main legal challenges related to the application of artificial intelligence in protected areas management? What legal measures can encourage the implementation of AI in sustainable forest management?” The responses provided by the experts from the three countries were analyzed and evaluated using content analysis, and the results are presented in Table 5 below:

Table 5. Legal challenges and legal measures related to the use of artificial intelligence in Turkey, Lithuania, and Morocco for protected areas management.

Category	Morocco	Lithuania	Turkey
Legal Arrangements	Inadequate legal arrangements for AI	Update existing AI regulations	AI integration into environmental laws
Data Privacy and Ethical Issues	Ethical concerns; data policies and concerns about changing jobs	GDPR compliance in data use; legal framework in privacy and audit	AI trustworthiness; ethical concerns and verification
Legal Measures	AI governance policies	Public–private partnership; current legal framework for AI	Legal promotion of AI use

Turkey, Lithuania, and Morocco have different advantages and disadvantages for the use of AI in the sustainable management of protected areas. Similarly, it is recognized that legal gaps for the use of AI in all three countries should be eliminated and ethical concerns should be eliminated. After the elimination of legal gaps and the elimination of ethical concerns in society, important steps will be taken towards the sustainability of protected areas, especially with international cooperation, and it is inevitable that the use of AI will become widespread in this process.

There are, of course, legal regulations on the environment in the country, but I think the effectiveness of these laws will increase with the widespread use of AI. The objectives targeted in the laws can be concretized with AI and can be used as an important tool for managers (LT3-4-9). One of the most important problems is environmental crimes. With the use of AI, detection and prevention of crimes can become easier and contribute to the fight against crime. Therefore, it can contribute to the sustainable management of protected areas (AT4-5). In Morocco, there is no specific legal framework for the use of AI in environmental management, which presents a challenge for the effective regulation and

implementation of AI technologies in this field (LM7-9, FEM1-8). Who is responsible for decision-making errors made by AI? If an AI system incorrectly detects an infringement in the natural environment or misses a significant event (e.g., illegal logging), it is necessary to clearly define the boundaries of responsibility (FML5-6, LL2-8).

The opinion is prominent in Lithuania and Morocco that specific AI regulations should be developed to fill legal gaps in these countries. In Turkey, however, it is evident that further legal frameworks need to be developed to improve environmental laws and enable more effective use of AI technologies.

Experts from all three countries were asked the following question: “Does the use of AI technologies in forest management create any challenges in terms of legal frameworks and ethical standards in Morocco? How is regulation and oversight applied in this?” The responses provided by the experts from the three countries were analyzed and evaluated using content analysis, and the results are presented in Table 6 below:

Table 6. Legal frameworks, ethical standards, and regulations in the use of artificial intelligence technologies in Turkey, Lithuania, and Morocco.

Country	Legal Challenges	Ethical Concerns	Oversight and Arrangements
Turkey	Lack of legal arrangements for AI	Data privacy and ethical concerns for AI	Lack of legal arrangements for AI and ethical concerns
Morocco	Lack of legal arrangements for AI Data privacy for AI	Questioning AI decision-making processes; data privacy and oversight	Lack of legal arrangements for AI Current Legal Arrangements
Lithuania	Lack of legal arrangements for AI Reliability of data generated by AI	AI validity and reliability Reliability of data generated by AI	Eligibility for EU Current Legal Arrangements

In Turkey, Lithuania, and Morocco, similar and different situations arise in terms of legal regulations and ethical concerns in the use of AI in the management of chromated areas. Similarly, legal gaps exist in all three countries for the use of AI in the management of protected areas. In Turkey, for example, there is a significant concern about the loss of personal data and its protection. In Morocco, there is no specific legal framework for the use of AI, but it is noted that regulations should be developed in the future. Lithuania, while aligned with EU AI regulations, lacks clear regulations on the use of AI in environmental monitoring and conservation.

There are uncertainties regarding the legal compliance of AI usage. In Turkey, there is a lack of concrete legal regulations for AI usage, and ethical standards are generally based on general technology laws (LT1-2-9). Legal regulations regarding the use of AI technologies have yet to be sufficiently developed. Ethical standards are still in the development phase. The absence of comprehensive legislation in this field complicates supervision and regulation (NGOET4-7-8).

The lack of legal arrangements and policies on AI is a major concern in Morocco. This is because it is often argued that data policies and data privacy may be violated. I think this is the most important obstacle to the use of AI (FEM7-8-9). The use of AI in the management of protected areas is inevitable once legal gaps are addressed. However, very serious regulations are required for this (FMM4-6).

In Lithuania, the use of AI in the nature conservation sector is not yet strictly regulated. However, the Lithuanian Environmental Protection Law provides a framework for AI usage in nature conservation measures and monitoring, offering a workable framework in this field (FML2, AL4-6).

While all three countries have potential for using AI in environmental management and nature conservation, they require a strong legal framework and supervision mechanisms to ensure the responsible use of these technologies.

Experts from all three countries were asked the following question: “What collaboration models or strategies would you suggest for public institutions to adopt AI technologies in forest management? What is the status and future of such collaborations?” The responses provided by the experts from the three countries were analyzed and evaluated using content analysis, and the results are presented in Table 7 below:

Table 7. The strategies for AI use in protected areas for Morocco, Lithuania, and Turkey.

Strategy Areas	Morocco	Lithuania	Turkey
Public–Private Partnerships	Encouraging collaboration with the private sector.	AI pilot projects and incubators for collaboration.	Stronger partnerships with the private sector.
Academic Research	Universities should lead AI-based environmental research.	Funding research centers for AI solutions.	Funding for academic research is provided.
International Cooperation	Partnerships with global organizations.	Participation in EU AI projects and experience exchange.	International collaborations and tech sharing.
Data Availability	Access to data for AI implementation.	Open data platforms for AI systems.	Open data policies should be developed.
Financial Support	-----	Tax breaks and innovation funds for AI projects.	Tax incentives and financial support are necessary.

Turkey, Lithuania, and Morocco are aware of current AI technology and are trying to create incentive policies for its use. Morocco is trying to take serious steps in this direction, encouraging academic research, supporting public–private sector cooperation, and emphasizing the importance of NGO participation.

In Morocco, the government prioritizes the Public-Private Partnerships (PPPs) model to encourage collaboration between public institutions and AI companies (FEM5-FEM2-8). Additionally, universities are encouraged to take leadership in AI-based environmental research projects, which can advance the integration of AI technologies in protected area management and environmental protection (FMM4-6).

In Lithuania, it appears that the country is working on establishing open data platforms to support the integration of AI technologies into protected area management strategies, while also incentivizing the private sector and academia through tax reductions and innovation funds. Furthermore, Lithuania values participation in European Union projects for international experience sharing. The Lithuanian government can promote the implementation of AI in forestry through clear strategies, financial support, and open data initiatives. By collaborating with business and academia, it can develop innovative solutions for sustainable forest management (FEL4-5-9).

In Turkey, the government aims to establish stronger collaborations with the private sector and academic institutions. Turkey believes that legal frameworks need to be developed for AI technologies, while also focusing on sector-specific training and raising awareness. Governments should collaborate with academia and the private sector to encourage the effective use of AI technologies in the forestry sector and implement regulations to ensure the applicability of these technologies (NGOET7-FET8). For AI to be used effectively in

protected area management, governments should create a strong legal infrastructure and encourage closer collaboration between the private sector and academia. Education and awareness-building are also important steps (LT1-2-7).

All three countries aim to establish open data sharing, financial incentives, and collaboration strategies to increase the use of AI. The importance of collaboration in the increased use of AI technologies stands out for all three countries. It is understood that AI technology usage will become more widespread through collaboration between the public and private sectors, as well as academia.

Experts from all three countries were asked the following question: “How do you see the role of AI technologies in protected areas management evolving in the future? What strategic plans does your country have to implement these technologies on a larger scale?” The responses provided by the experts from the three countries were analyzed and evaluated using content analysis, and the results are presented in Table 8 below:

Table 8. The future role of AI technologies and the strategic plans to implement them on a larger scale.

Themes	Morocco	Lithuania	Turkey
Future AI Implementation Plans	Updating environmental policies for AI. Supporting the use of AI in research.	AI Action Plan 2023–2026. Forest 4.0 project. Academic initiatives for AI.	Legal regulations for AI technologies. Investment in AI research.
Scaling AI in Protected Areas Management	Expanding AI in national parks. Partnerships with academia and tech firms. Policy-making support for sustainability.	Forest data management systems. Collaborations between academia and government. Expansion of AI applications.	Stronger AI integration with local authorities. Collaboration between government, local authorities, and academia. Strengthening digital infrastructure.
Strategic Plans	Investment and government policies for AI. Enhancing monitoring and fire prevention.	AI-driven forest management solutions. Integration of AI into national forest management policies. Innovations in AI for forest sustainability.	Inadequate legal arrangements for AI. Government policies for AI. AI-powered research.

In all three countries surveyed, there is a growing trend towards the use of AI in the management of protected areas. However, expectations and purposes of use differ. For example, in Turkey, priority in the use of AI is given to combating forest fires, as well as monitoring environmental health. Finally, especially in Turkey, academia’s interest in this field is increasing, and private sector investments are becoming more widespread.

There is significant potential for AI technologies to be used in forest management. Turkey’s strategic plans in this area should focus particularly on monitoring forest fires and protecting biodiversity (FMT7-9, LT1-3). Academia can provide guidance in the implementation of these technologies. The integration of AI technologies into forest management could revolutionize data collection and analysis. For the successful implementation of these technologies in Turkey, local governments and the state need to establish strategic partnerships (NGOET4-FET9). AI tools can be highly effective in the management of protected areas as they work with real-time data, but the correct needs must be identified (AT-7). In Turkey, the use of AI technologies should be increased, particularly in combating forest fires, and their proper application for early detection should be widely promoted (NGOET-4).

Morocco, on the other hand, aims to use AI technologies as a tool for environmental protection and sustainable forest management, and it has initiated projects to develop AI-based research and decision support systems. In Morocco, it is suggested that the government needs to increase sectoral collaborations and pilot projects to expand the use of these technologies.

Morocco's future for AI in forest management focus on integrating AI into national environmental policies, increasing investment in AI-driven research and monitoring systems, and developing decision-making frameworks for conservation (AM7-8). The country aims to scale AI by expanding pilot projects to national parks and strengthening partnerships with academia and technology firms (LM4). Additionally, Morocco is working on encouraging AI-based policymaking to support sustainability goals, fostering a collaborative approach to enhance forest management (NGOEM5).

Lithuania, meanwhile, aims to use AI technologies in areas such as data collection and forest health monitoring, having developed an AI action plan for 2023–2026, and is working to strengthen academic collaborations through innovative projects like Forest 4.0.

The Forest 4.0 project aims to create a center of excellence for forest-related AI and Internet of Things (IoT) technologies, involving Vytautas Magnus University, Linnaeus University (Sweden), Kaunas University of Technology, and companies Art21 and Agrifood (FEL5). The Ministry of Environment emphasizes the need to improve the efficiency of forest information collection, automate data collection and monitoring, and implement AI solutions (AL5-7).

All three countries acknowledge the potential of AI technologies to make forest management more efficient, but their strategic plans and implementation methods are shaped according to the specific conditions of each country. With developing technological advances, Turkey, Morocco, and Lithuania appear eager to integrate AI applications and tools into protected area management. Particularly, government support, legal regulations, and collaborative approaches stand out as key aspects.

4. Discussion

Turkey, Lithuania, and Morocco each possess significant natural resources and green spaces, forming an important part of their shared heritage. In all three countries, efforts to sustainably manage and protect protected areas stand out. However, this process requires an integrated approach that aligns with technological advancements and is rooted in accuracy. Particularly, high human mobility in protected areas reveals several issues. Specifically, insufficient legal regulations and local resistance to these regulations exacerbate these problems, hindering the desire for sustainable management.

Protected areas are characterized as areas where people in industrial society can escape from the negative effects of intensive urbanization and be alone with nature, while at the same time creating natural environments for biodiversity and the continuity of species [52]. However, they are critical for carbon dioxide mitigation and play a key role in tackling the global climate change challenge [53].

The research findings are organized under three main headings, with themes shaped within these categories based on data analysis. These headings are: “legal challenges”, “technology adoption and its barriers”, and “collaboration strategies”.

(a) Legal Challenges

The most prominent issue in all three countries is the lack or inadequacy of legal regulations encountered in the process of widespread adoption of AI technologies. While there are laws and regulations aimed at the protection of the natural environment in all

three countries, the opinion that the enforcement of these laws and sanctions is weak is prevalent. Additionally, the preparation of new and updated legal regulations to promote the use of AI technologies is under consideration.

In recent times, issues directly related to climate change, such as forest fires, droughts, water scarcity, and food security, have become frequent topics of discussion in Turkey. Strengthening legal measures to address these issues has become inevitable. Updating and tightening the legal and institutional framework for forested areas and protected regions is essential. The role and importance of legal structures in solving environmental problems are increasingly significant [54–56]. The most significant legal foundation for achieving environmental sustainability in Turkey is the 1982 Constitution. Article 56 emphasizes the fundamental human rights to a healthy environment and health [56], highlighting the importance of involving citizens in this process. Furthermore, Article 44 of the Constitution, which refers to the efficient use, protection, and development of lands, establishes constitutional responsibilities [57].

The protection and sustainability of protected areas and forested regions require robust institutional structures and comprehensive legal regulations. In particular, the monitoring of ecosystems and the conservation of biodiversity are directly linked to these strong institutional and legal frameworks. Individual efforts, while valuable, often prove insufficient over time and may pose risks to sustainability goals. Therefore, decisive institutional actions coupled with stringent legal measures can be regarded as the most critical steps toward achieving environmental sustainability [58–60].

Protecting natural areas requires not only strong legal measures but also public support and participation. Following the rules and accepting legal actions help ensure their sustainable management.

The insufficient involvement of local communities in the sustainable management and decision-making processes of protected areas can be regarded as one of the most critical challenges encountered in this context. Addressing this issue necessitates the integration and utilization of AI technologies in a manner that also fosters public participation. In this way, AI-supported decision-making mechanisms may contribute to a more participatory governance model. In line with this perspective, Wang and Zhang found that the use of AI technologies in Chinese tourism SMEs led to increased customer engagement. This, in turn, enhanced the enterprises' environmental, social, and governance (ESG) performance. Therefore, public participation should not be overlooked during the technological integration process, and individuals and groups must be actively included in the management of protected areas [61].

In Lithuania, the protection and management of protected areas are ensured through various legal frameworks, including *Forest Law No. IX-240*, *Law No. I-301 on Protected Areas*, and *Law No. I-2223 on Environmental Protection*. These laws provide the necessary legal framework for the sustainable use of protected areas by the state. However, in a developing and evolving world, these laws have proven to be insufficient, and more stringent measures are needed. Whether existing legal regulations are sufficient for a sustainable management approach is constantly debated and initiatives to update them are ongoing [62–64].

One of the main barriers to the use and widespread adoption of AI technologies in Turkey, Lithuania, and Morocco is insufficient legal regulations and their implementation. Directly related legal regulations concerning the use of AI technologies for the sustainable management of protected areas need to be urgently enacted by governments [65]. In this regard, the European Union's AI Act serves as an important example, containing regulations related to the use of AI [66]. In all three countries, it is crucial to address the legal gaps regarding the use of AI for the sustainable management of protected areas. It

can also be argued that opportunities for technology transfer should be considered for the protection of wildlife [63].

AI technologies are frequently utilized across all sectors. The use of AI technologies in the management of forested areas and protected regions is also inevitable. However, it is necessary to establish a legal infrastructure in this process. The weak legal foundations in each of the three countries make this issue particularly important. Greater emphasis should be placed on legal regulations concerning AI technologies, and legal challenges should be addressed to facilitate their widespread adoption [67,68].

In all three countries studied, inadequate legal regulations and political gaps in the management processes of protected areas are considered as important obstacles. By removing these obstacles and making the necessary legal arrangements, concrete steps can be taken for sustainable environmental goals, and the protection of biodiversity can be ensured. Additionally, increasing investments in the infrastructure necessary for the protection of these areas can also contribute to sustainable management processes.

(b) Technological Barriers

The second prominent issue/barrier in all three countries is the willingness to accept and use AI technologies, particularly among local communities and managers. AI technologies are currently used in many sectors. They can offer unique opportunities for the sustainable management of protected areas. AI technologies can contribute to various fields such as ecosystem conservation, biodiversity monitoring, and environmental protection, providing valuable support for sustainable management processes. However, in all three countries, there is a need for actions such as developing technological infrastructure and allocating sufficient financial resources for this purpose.

In the case of Turkey and Morocco, artificial intelligence applications are used in the sustainable management of protected areas for the detection and prevention of forest fires, identification of illegal activities, and integration into early detection systems. Although artificial intelligence (AI) technologies facilitate the monitoring of illegal activities and support management processes, they may also constitute a fundamental source of concern. To mitigate these concerns and minimize potential errors, several key measures can be implemented [69]. First, ensuring the availability and proper processing of high-quality data is essential. This involves the systematic categorization and refinement of datasets to improve model accuracy. Second, AI tools and software must be regularly updated to reflect evolving threats and environmental changes. Third, the adoption of multi-model systems within AI applications is recommended; by combining algorithms with varying strengths, classification errors—such as false positives—can be significantly reduced. Fourth, integrating human expertise into AI-based decision-making processes (i.e., human-in-the-loop systems) ensures that questionable outputs undergo validation and quality control. Finally, continuous monitoring and performance evaluation of deployed models are necessary, along with periodic updates based on real-world data feedback. Adhering to these steps enhances the reliability and effectiveness of AI tools in supporting sustainable and error-resilient management strategies [70,71].

The technological infrastructure deficiencies of countries are highly determinative in the use of AI technologies in protected areas, leading to their limited application. Particularly, as each of these countries falls within the category of developing nations, it is evident that robust infrastructures are needed for the widespread adoption of AI technologies [72]. For instance, the collection of large datasets, their storage, and reliable analysis require significant and strong infrastructure, as well as technology transfer initiatives. This is especially critical since protected areas may involve complex and variable datasets [73]. To overcome these barriers, a strong digital infrastructure must be established, and technical training at the local level should be provided.

In Lithuania, however, it appears that AI applications have not yet been fully integrated into management processes. Artificial intelligence applications analyze complex data using various methods, making it more understandable and evolving through the experiences it gains [74]. These systems are employed in various sectors, one of which is the management of forested areas and protected regions and are being rapidly integrated.

Modern technology is increasingly integrating artificial intelligence (AI) into all management processes, and its usage is becoming more widespread [75–77]. One of the most prominent issues encountered in the management of protected areas is forest fires. These fires present a significant risk factor and pose a serious threat to biodiversity [78,79]. In reducing these risks, AI-based applications, and tools, such as image processing in areas like fire prevention, detection, land recognition, and real-time monitoring of the fire process, are being explored. This is because AI-based autonomous systems used in fire-fighting efforts are known to be quite significant [80–82]. Although AI technologies minimize time and damage, several reasons explain why these technologies have not yet been used in Lithuania. A slow adoption of digital technologies in the management of protected areas stands out. Moreover, the sector's cautious approach to innovations makes the integration of advanced technologies like AI more difficult. Additionally, legal and governance barriers, along with hesitations about data sharing among stakeholders, increase concerns about trust and privacy in the use of digital technologies [83,84].

The use of AI in the sustainable management of protected areas is becoming more common, but it also brings about legal and ethical concerns. Legal apprehensions and ethical concerns regarding the use of AI are prominent in the three countries under study.

Lack of transparency in decision-making processes, algorithmic biases, and data privacy issues are primary ethical concerns that need attention in forestry practices [85,86]. Similar concerns are also raised in the international literature. For instance, the European Union's AI Act emphasizes that AI applications must be assessed within an ethical and legal framework. This law aims to ensure that AI is used in a safe, transparent, and accountable manner [87]. While the European Union has implemented legal regulation regarding AI, there is no such legislation in Turkey, Lithuania, and Morocco. To address the legal and ethical concerns mentioned above, it can be argued that all three countries need to create legal regulations regarding the use of AI.

The adoption of AI technologies by managers and their integration into management processes is particularly contingent upon the elimination of legal uncertainties. When individuals responsible for the management of protected areas can operate with confidence in established legal frameworks, they are more likely to make stronger and more decisive decisions [88]. The removal of legal ambiguities will enhance managers' willingness to assume responsibility, and sustainable practices supported by technology transfer will yield more effective outcomes.

Issues such as data ownership, lack of transparency, algorithmic bias, and data privacy have raised serious concerns during the implementation of AI technologies. Moreover, the inadequacy of regulations regarding informed consent, confidentiality, and data security in the process of collecting data from local communities poses significant ethical challenges. Such concerns contribute to a growing distrust of AI technologies and negatively influence their acceptance and adoption [89]. Additionally, algorithmic biases inherent in AI systems may lead to the marginalization of both natural elements and local communities, thus undermining the inclusivity of decision-making processes [90]. Therefore, for AI applications in protected areas to be effectively implemented, it is essential not only to ensure technical competency but also to establish solid ethical and legal foundations [91]. Otherwise, due to legal and ethical concerns, administrators will be reluctant to adopt AI.

(c) Collaboration Strategies

The third main issue regarding the use of AI technologies in Turkey, Lithuania, and Morocco is the weakness of public–private sector collaborations. The involvement of all institutions and organizations in these countries within the framework of sustainable environmental management can influence the use of AI technologies. Specifically, with the inclusion of the private sector by governments, the workload of public administration is expected to be reduced. Furthermore, the support of local communities in the process and their assistance in inter-institutional collaborations can also be decisive. In all three countries, the need for collaboration between the public and private sectors appears to be crucial for the sustainable management of protected areas.

In Turkey, Lithuania, and Morocco, alongside addressing the legal deficiencies in the use of AI in the sustainable management of protected areas, public–private collaboration and academic involvement are significant factors. Specifically, government incentives, tax reductions, and support for innovative solutions are expected to pave the way for the use of AI in the sustainable management of protected areas.

Collaboration between the public and private sectors in the widespread use of AI can enhance the societal benefits of technology transfer [92]. The 1st Workshop Report of the Turkey AI Initiative highlights the importance of fostering close collaborations between academia, the private sector, and the public to advance AI research. Such collaborations can contribute to identifying priority research areas [93]. Especially, green supply chain options have gained significant importance, particularly in inter-organizational supply chain processes, and enhancing inter-organizational cooperation is crucial for sustainable business management. It is vital to expand these collaborations not only among public institutions but also between the private sector, civil society organizations, and other entities. Therefore, increasing inter-organizational cooperation could accelerate the integration of AI technologies into management processes [94].

The importance of these collaborations is increased as protected areas are among the primary and sustainable areas of focus. The effective and ethical proliferation of AI technologies can be said to rely on public–private collaboration and joint academic efforts. This tripartite structure plays a critical role in the development, implementation, and regulation of AI applications, thereby supporting the technology's societal benefits [95–97].

Public–private sector collaboration is of paramount importance in the adoption and widespread use of AI technologies. Particularly, with increased public support, private sector representatives are more likely to invest in this field. Consequently, the development, updating, and broader implementation of AI technologies can be anticipated. Moreover, the societal acceptance of such partnerships can enhance social benefits and outputs related to environmental sustainability [98]. In an era where technology transfer is accelerating, collaborations between the public and private sectors are becoming increasingly crucial.

Public–private collaborations are essential for integrating AI applications into sustainable management practices, making them an inevitable tool for inclusive management. Furthermore, it is expected that through government incentives, support, and collaborations, a combination of technical innovations and market-oriented approaches will help address complex societal issues [99,100]. The World Economic Forum highlights that, through collaboration and public support, AI applications will play a significant role across all sectors of society and in management processes [101]. The intervention of governments plays a crucial and decisive role in enhancing the efficiency and productivity of AI technologies. This, in turn, facilitates the transfer of technology and the development of an innovative mindset. The management processes of protected areas can be improved not only through financial support but also through collaboration and coordination [102]. The widespread adoption of socially inclusive public–private partnerships positively influences the accep-

tance of technology, offering new opportunities for sustainable management models [103]. Encouraging stakeholder participation and ensuring the acceptance of government interventions within society can not only enhance the effectiveness of AI technologies but also accelerate the sustainable management of protected areas.

For AI technologies to gain societal acceptance, their dissemination across broad segments of society and support through inter-institutional collaboration are essential. The convergence of protected area management and the technology sector can be achieved through local partnerships. In particular, public-sector leadership plays a critical role in facilitating the broader adoption of AI technologies by engaging other sectors in the process [104]. Governments can foster incentives by supporting stakeholders in the private sector and can take the lead in generating AI-based environmental projects. Furthermore, involving local communities in collaborative processes may enhance the acceptance of technology, thereby fostering broad societal consensus for environmental sustainability [105].

The use of AI in the sustainable management of protected areas in Turkey, Lithuania, and Morocco holds great potential, depending on the creation of necessary legal frameworks and infrastructure. The sustainable management of protected areas is vital for combating climate change, biodiversity loss, and deforestation. Human-induced stresses require innovative approaches to ensure the long-term health and productivity of forest ecosystems [106]. Fire prevention, biodiversity conservation, and forest health maintenance are critical determinants. Kinaneva et al. emphasized that traditional methods for fire management are both hazardous and costly [107]. In this context, it is envisaged to activate smoke detectors with automatic detection capabilities in the fight against forest fires, and attention has been paid to the fact that this is AI-based. In addition, the establishment of automatic surveillance systems aims to eliminate possible forest fire threats [108].

The use of AI technologies provides the opportunity to work with large datasets, which in turn provides an opportunity to protect the ecosystem within protected areas. For example, remote monitoring and tracking systems with AI technology facilitate the regional detection of forest fires and offer early intervention opportunities [109]. However, considerable attention must be given to the technical infrastructure and operational processes when utilizing these AI tools. For instance, seamless integration between various AI platforms and systems is essential for sustainable management processes [110]. This requires eliminating barriers to data exchange between heterogeneous AI tools and ensuring that all systems are fully operational. Middleware architectures and API standardization can be crucial tools in this regard. In this way, multiple AI tools can operate in a coordinated manner, contributing effectively to management processes. The primary objective should be to ensure that different data sources work in harmony to achieve maximum efficiency [111,112].

The use of AI technologies is of course not limited to combating forest fires. It also plays an active role in monitoring and protecting environmental health and protecting biodiversity. There are also areas of use that make a difference, such as natural resource management and the use of sustainable methods for these resources. Management processes supported by artificial intelligence (AI) technologies are capable of analyzing complex environmental dynamics associated with time- and space-dependent internal and boundary sources, thereby enabling the development of analytical solutions [113]. Therefore, the use of AI in protected areas is important for environmental sustainability and also contributes to the development of a sustainable management approach. This situation creates a more advantageous management approach compared to traditional methods [114]. These technologies are of vital importance in conserving ecosystems and mitigating the effects of climate change by enabling real-time monitoring, optimizing resource allocation, and making data-driven decisions [106].

It has an important role in reducing carbon dioxide and enables the effective and efficient management of natural resources. It has a critical importance in combating global climate change within environmental sustainability goals. In addition, it can contribute to the protection of natural life, elimination of biodiversity risks and provision of a healthy environment [115,116]. In this context, they help ensure the efficient use of energy and support efforts to maintain ecological stability. By encouraging strategies that aim to reduce environmental harm and use resources more wisely, they form a key part of global efforts to address climate challenges [117,118]. In the long run, their use in policies and practical applications can strengthen environmental resilience and support goals such as those set by the United Nations for sustainable development [119]. The implementation and adoption of AI technologies in protected areas are of critical importance in the context of sustainable development goals. In particular, achieving environmental sustainability objectives necessitates not only addressing the human factor but also integrating technological applications into management processes. By supporting the decision-making processes of managers responsible for protected areas, the mission of environmental sustainability outlined within the United Nations Sustainable Development Goals can be effectively advanced [120,121].

The research findings focus on the acceptance theories of AI technologies in the sustainable management of protected areas in Turkey, Lithuania, and Morocco. In particular, the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) provide a meaningful theoretical foundation for environmental conservation and sustainable management practices. TAM posits that the attitudes of individuals and societies towards new and unfamiliar AI technologies are determinants of their acceptance [122–124]. Therefore, the research findings highlight and strongly emphasize the adoption of AI technologies in the sustainable management of protected areas. Furthermore, UTAUT identifies four key factors that determine the use of new technologies: performance expectancy, effort expectancy, social influence, and facilitating conditions. These factors overlap with aspects such as trust in AI technologies, shared understanding of technology within the community, and infrastructure limitations [125,126].

From an institutional perspective, the theoretical framework for the acceptance and use of AI technologies is directly related to the research findings. It is also closely connected to how institutional structures, policy-making processes, and collaboration mechanisms are shaped [127]. In this context, the lack of resistance from local communities in the management processes and their participation, as well as strong public–private sector collaborations, contribute to the widespread use of AI technologies in protected areas [99]. Moreover, collaborative governance theory emphasizes the effectiveness of stakeholders from different sectors working together around common goals in achieving set objectives [128]. In this regard, increasing technology acceptance, improving attitudes towards new applications and technologies, and expanding collaborations can significantly contribute to the sustainable management of protected areas.

5. Conclusions

In this study, a total of one hundred forty-five (145) experts from Turkey, Lithuania, and Morocco were interviewed. These experts included lawyers with experience in environmental issues, academics working in the relevant field, forest management officials from protected areas, government forestry experts, and representatives from non-governmental organizations. Semi-structured interview forms were used, and interviews were conducted with one hundred forty-five (145) experts in total from the three countries. According to the findings, the following is derived:

- ✓ In Turkey and Lithuania, the main issues faced in protected areas are the inadequacy of legal regulations, resistance from local communities toward legal processes, and

lack of participation, whereas in Morocco, infrastructure deficiencies and illegal activities are the primary challenges;

- ✓ In the management of protected areas, similar themes emerged regarding the use of AI in all three countries. The most significant barrier to AI use was the lack of legal regulations. However, it was also noted that increasing infrastructure investments and enhancing collaboration at the local level would promote the wider adoption of AI;
- ✓ When evaluating the impact of AI use on the sustainable management of protected areas, similar themes emerged in all three countries. Particularly, forest fire detection and suppression, biodiversity conservation, prevention of illegal activities, and early detection systems were identified as areas where AI could significantly improve the management of protected areas;
- ✓ A key finding highlighted that for the widespread adoption of AI in the sustainable management of protected areas, public–private sector collaborations should be enhanced in all three countries. Additionally, governments should provide tax reductions and develop incentives for innovative applications, and academia should offer scientific data support in this field.

In all three countries, important and striking conclusions have been reached about the spread and future of the use of AI technologies:

- ✓ In the fight against forest fires, prevention of illegal activities, environmental sustainability, conservation of natural resources and combating climate change, it is inevitable to integrate technology transfer into the sustainable management of protected areas. For this, governments in all three countries are expected to support and encourage managers through tax breaks and incentive policies.

Government incentives and tax breaks alone may not be sufficient for the diffusion of AI technologies. According to experts from Turkey, Lithuania, and Morocco, in all three countries the following points are made:

Training programs and project-based initiatives should be implemented to increase technological literacy. Measures should be taken to improve the level of acceptance and use of technology among protected area managers. Community participation should also be encouraged to raise environmental awareness. Furthermore, allocating more resources from public budgets could enable the implementation of pilot projects in specific regions in the countries concerned. Artificial intelligence applications, which are expected to be a critical part of the process of environmental sustainability and combating climate change, should urgently be integrated into management processes to support managers' decision-making and action.

This research focuses on the applicability of AI technologies in the sustainable management of protected areas in Turkey, Lithuania, and Morocco, identifying the barriers to the use of these technologies. The findings highlight legal inadequacies, infrastructure limitations, and the need for public–private sector collaboration in all three countries. This study also provides practical recommendations for policymakers and managers. The comparative analysis conducted across Turkey, Lithuania, and Morocco points to the presence of similar challenges in all three countries.

This research is based solely on qualitative data. The rationale for selecting the three countries is detailed in the Methodology section. However, future studies are expected to include a broader range of countries, supported by both qualitative and quantitative data. The geographic scope being limited to three countries and the AI applications being addressed only in general terms also present opportunities for future research. In this regard, future studies involving larger datasets, including technical analyses and

encompassing different regional contexts, would further enrich the literature on the use of AI in environmental sustainability.

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